

DATA STRUCTURE

LAB PROGRAMS

6. Write a C program to find Fibonacci series using Recursion

```
#include <stdio.h>
```

```
int fibonacci(int n) {
```

```
    if (n == 0)
```

```
        return 0;
```

```
    else if (n == 1)
```

```
        return 1;
```

```
    else
```

```
        return fibonacci(n - 1) + fibonacci(n - 2);
```

```
}
```

```
int main() {
```

```
    int n, i;
```

```
    printf("Enter the number of terms: ");
```

```
    scanf("%d", &n);
```

```
    printf("Fibonacci Series:\n");
```

```
    for (i = 0; i < n; i++) {
```

```
        printf("%d ", fibonacci(i));
```

```
    }
```

```
    return 0;  
}
```

```
Enter the number of terms: 9  
Fibonacci Series:  
0 1 1 2 3 5 8 13 21  
-----  
Process exited after 1.993 seconds with return value 0  
Press any key to continue . . .
```